higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MULTIMEDIA BASICS
NQF LEVEL 2

(10040162)

8 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 10 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.15) in the ANSWER BOOK.

1.1.1 Which ONE of the following is NOT a type of story plot?

A Story of my life
B Comedy
C Tragedy
D Rebirth

1.1.2 The … is the most important component of an entertainment story.

A protagonist
B antagonist
C conflict
D conclusion

1.1.3 A stand-alone episode of a television series that is used to sell the show to a television network is called a … episode.

A final
B pilot
C soapie
D movie

1.1.4 Character development in storytelling involves ...

A planning the character's appearance and history.
B making the character the antagonist.
C making sure the character appear more than anybody else in the story.
D giving the character a key role in the story.

1.1.5 A struggle taking place inside the character's mind is called …

A confusion.
B a clash of ideas.
C internal inspection.
D internal conflict.

1.1.6 The protagonist in a story is usually …

A the good guy in the story.
B a little-known character in the story.
C a friend of the main character.
D an incidental character in the story.

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1.1.7 The time and location in which a story takes place is called a/an ...

A stage.
B theatre.
C acting studio.
D setting.

1.1.8 The sequence of events in a story is a ...

A story line.
B story plan.
C plot.
D body.

1.1.9 A/An ... is an account of imaginary or real people and events told for entertainment.

A episode
B story
C series
D event

1.1.10 The outline of the plot of a play, film or book is called a ...

A synopsis.
B storyboard.
C conclusion.
D introduction.

1.1.11 The processes of making one image conform to the size, shape, and/or texture of another image is referred to as ...

A tweening.
B mapping.
C animation.
D blending.

1.1.12 The process by which a computer model of a 3D-object is turned into a real object is called ...

A drawing.
B sketching.
C rapid prototyping.
D tweening.

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1.1.13 ... light exists in nature, for example sunlight or moonlight, and depends on the time of day.

A Natural
B Incident
C Reflected
D White

1.1.14 ... is the final process of fine-tuning and tweaking the mix of a song or collection of songs from which the master will be created.

A Editing
B Mastering
C Mixing
D Filtering

1.1.15 The ... is a transducer which is used to convert sound pressure waves into electrical signals.

A speaker
B tweeter
C bass
D microphone

(15 × 1) (15)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.15) in the ANSWER BOOK.

1.2.1 A storyboard is the same as the script of a story.

1.2.2 Vector graphics are saved as mathematic calculations.

1.2.3 A film story is usually longer than a television story.

1.2.4 An autobiography is an example of a nonfictional story.

1.2.5 Animation added to a video usually creates a sense of interest.

1.2.6 Path animation is an example of 2D-animation.

1.2.7 The higher the peaks of sound waves, the softer the sound.

1.2.8 A 16-bit image occupies less storage space on a computer than an 8-bit image.

1.2.9 Bitmap images can be stretched as much as you want without distorting their appearance.

1.2.10 The word e-zines refers to magazines published electronically on the internet.

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1.2.11 Leading refers to the distance between lines of text.

1.2.12 One needs a licence to use open-source software.

1.2.13 Daily newspapers can be bought and read on-line.

1.2.14 Decorative typeface is suitable for paragraphs in print material.

1.2.15 What is seen on a digital screen is a result of light distribution.

(15 × 1) (15)

1.3 Complete the following sentences by using the words in the list below. Write only the word or words next to the question number (1.3.1–1.3.15) in the ANSWER BOOK.

pictograph; sans serif; serif; kerning; negative space; luminance; symbol; emphasis; balance; texture; line; plane; horizontal lines; volume; tranquility

1.3.1 A … is a flat surface.

1.3.2 The amount of space occupied by an object measured in three dimensions and expressed in cubic units is called …

1.3.3 A ... is a mark or character used as a conventional representation of an object, function or process.

1.3.4 … is the space around and between the subjects of an image.

1.3.5 ... is the intensity of light emitted from a surface per unit area in a given direction.

1.3.6 A small decorative line added as embellishment to the basic form of a character is called a …

1.3.7 ... refers to the distribution of visual weight in a work of art.

1.3.8 The visual feel of a two-dimensional work in art design is called ...

1.3.9 ... suggests a feeling of rest, relaxation or restfulness.

1.3.10 ... could be defined as an area or object in the artwork which draws attention and becomes a focal point.

1.3.11 Letter forms which do not have extending features at the end of strokes are known as ...

1.3.12 ... increases or decreases the spacing between certain pairs of letters.

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1.3.13 A ... is a long mark or stroke made on a surface with a pen, pencil or tool.

1.3.14 ... refers to an untroubled state free from disturbances.

1.3.15 A graphic representation of statistics on a chart, graph or computer screen is known as a ...

(15 × 1) (15)

1.4 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–F) next to the question number (1.4.1–1.4.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.4.1	CODEC	A	technique of projecting a 2D-image onto 3D-geometry
1.4.2	Frame	B	process of projecting a two-dimensional image onto a three-dimensional surface
1.4.3	Projection map	C	total pixel size of a bitmap image
1.4.4	Texture mapping	D	first renders before final compositing
1.4.5	Resolution	E	compressor/decompressor
		F	still image in animation and the basic unit of time measurement

(5 × 1) (5)

[50]

TOTAL SECTION A: 50

SECTION B

QUESTION 2

2.1 Briefly explain each of the following types of animation:

2.1.1 2D-animation or cel animation

2.1.2 3D-computer animation

2.1.3 Motion graphics

2.1.4 Stop motion

(4 × 2) (8)

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2.2 Define *web animation.* (2)

2.3 Describe the process of motion capturing. (2)

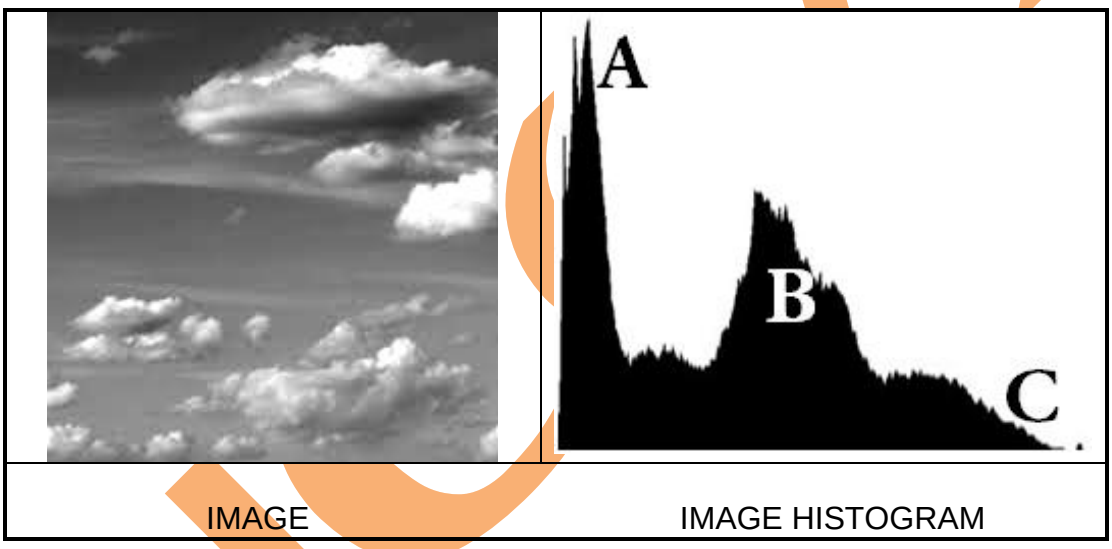
2.4 List FIVE types of lights that could be used in 3D-animation projects. (5)

2.5 Name THREE pieces of hardware equipment needed to record voice. (3)
[20]

QUESTION 3

3.1 List THREE software animation programs that could be used for animation projects. (3)

3.2 Study the image and histogram of a cloudy, grey sky.



3.2.1 Discuss the areas labelled A, B and C on the histogram with relation to the image. (3 × 2) (6)

3.2.2 Which software tool can be used to make this image brighter? (2)

3.2.3 What is the purpose of filters in image processing? (2)

3.3 Name ONE peripheral device used for multimedia sound output. (1)

3.4 Name SIX sequential steps in the production process of 3D-graphics. (6)
[20]

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QUESTION 4

4.1 Explain TWO types of 3D-surface mapping. (2 × 3) (6)

4.2 Describe TWO differences between music saved on a *cassette* and on a *USB.* (2 + 2) (4)

4.3 Discuss any THREE image altering or editing actions that could be used on a photograph to improve or enhance the image. (3 × 2) (6)

4.4 Describe the main differences between the RGB and CMYK colour models. (2 × 2) (4)

[20]

QUESTION 5

5.1 Name THREE different types of camera lenses. (3)

5.2 Explain the use of the following photography items:

5.2.1 Graduated filter

5.2.2 Neutral density filter

5.2.3 UV filter

5.2.4 Built-in flash

5.2.5 Camera stand

(5 × 2) (10)

5.3 Explain each of the following camera angles:

5.3.1 High camera angle

5.3.2 Low camera angle

(2 × 2) (4)

5.4 Name THREE types of noises in audio recordings. (3)

[20]

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QUESTION 6

6.1 Define each of the following with regard to entertainment stories:

6.1.1 Antagonist

6.1.2 Hero

(2 × 2) (4)

6.2 Explain each of the following sound effects:

6.2.1 Echo

6.2.2 Delay

6.2.3 Reverb

6.2.4 Distortion

(4 × 2) (8)

6.3 Discuss the difference between the *tempo* and the *pitch* of music. (2 × 2) (4)

6.4 List any THREE text file saving formats in multimedia. (3)

6.5 What is the purpose of the image cropping tool? (1)

[20]

TOTAL SECTION B: 100
GRAND TOTAL: 150

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